

REDI-KLEEN II
Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 力滌淨第二代通用清潔劑
REDI-KLEEN II

Other means of identification : Not applicable.

Recommended use : Cleaning product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 1.0 % - 5.0 %

Company : Ecolab Limited
15/F, Lu Plaza, 2 Wing Yip Street
Kwun Tong, Kowloon Hong Kong
+852-2372 7768

Emergency telephone : 800-968-793 (Hong Kong)
number : 1-651-222-5352 (United States)

Issuing date : 2021.09.24

Section: 2. HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Acute toxicity (Oral) : Category 5
Skin corrosion/irritation : Category 1B
Serious eye damage/eye : Category 1
irritation

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS Label element
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : May be harmful if swallowed.
Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**
Wash skin thoroughly after handling. Wear protective gloves/
protective clothing/ eye protection/ face protection.
Response:
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN
(or hair): Take off immediately all contaminated clothing. Rinse skin
with water/shower. IF INHALED: Remove person to fresh air and keep
comfortable for breathing. Immediately call a POISON

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician. Call a POISON CENTER or doctor/ physician if you feel unwell. Wash contaminated clothing before reuse.

Storage:

Store locked up.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Precautionary Statements

Prevention:

Wash hands thoroughly after handling.

Response:

Get medical advice/ attention if you feel unwell.

Storage:

Store in accordance with local regulations.

Product AS SOLD

Other hazards

: None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS**Product AS SOLD**

Pure substance/mixture

: Mixture

Chemical Name**CAS-No.****Concentration: (%)**

Sodium Xylenesulfonate

1300-72-7

5 - 10

Inorganic silicate.

1312-76-1

1 - 5

2-butoxyethanol

111-76-2

1 - 5

Potassium Dodecylbenzenesulphonate

27177-77-1

1 - 5

potassium hydroxide

1310-58-3

1 - 5

Sodium silicate

1344-09-8

1 - 5

Product AT USE DILUTION

No hazardous ingredients

Section: 4. FIRST AID MEASURES**Product AS SOLD**

In case of eye contact

: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact

: Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed

: Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled

: Remove to fresh air. Treat symptomatically. Get medical attention.

Protection of first-aiders

: If potential for exposure exists refer to Section 8 for specific personal protective equipment.

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

Section: 5. FIREFIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during firefighting	: Exposure to decomposition products may be a hazard to health.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides metal oxides Sulphur oxides
Special protective equipment for firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Section: 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 40 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
2-butoxyethanol	111-76-2	OEL-TWA	20 ppm 97 mg/m ³	HK OEL
potassium hydroxide	1310-58-3	OEL-C	2 mg/m ³	HK OEL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Colour	: clear, light yellow	light yellow
Odour	: slight	odourless
pH	: 13.0 - 14.0, (100 %)	11.0 - 13.0
Flash point	: Not applicable.	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: no data available	
Flammability (solid, gas)	: Not applicable.	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	
Vapour pressure	: no data available	
Relative vapour density	: no data available	
Relative density	: 1.07 - 1.12	

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Water solubility	: soluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: The substance or mixture is not classified as oxidizing.
Molecular weight	: no data available
VOC	: no data available

Section: 10. STABILITY AND REACTIVITY

Product AS SOLD

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: Acids Metals
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides metal oxides Sulphur oxides

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
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MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion, Irritation
Skin contact	: Redness, Pain, Irritation, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity	: Acute toxicity estimate : 4,799 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : > 40 mg/l Test atmosphere: vapour
Acute dermal toxicity	: Acute toxicity estimate : > 5,000 mg/kg
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: no data available
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available
Germ cell mutagenicity	: no data available
Teratogenicity	: no data available
STOT - single exposure	: no data available
STOT - repeated exposure	: no data available
Aspiration toxicity	: no data available

Section: 12. ECOLOGICAL INFORMATION

Toxicity

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Product AS SOLD

Environmental Effects : This product has no known ecotoxicological effects.

Product AT USE DILUTION

Environmental Effects : This product has no known ecotoxicological effects.

Product AS SOLD

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Inorganic silicate.
96 h LC50: 210 mg/l

2-butoxyethanol
96 h LC50: 1,474 mg/l

Potassium Dodecylbenzenesulphonate
96 h LC50 Fish: 3.2 mg/l

Sodium silicate
96 h LC50 Oncorhynchus mykiss (rainbow trout): 260 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : 2-butoxyethanol
48 h EC50: 690 mg/l

Sodium silicate
48 h EC50 Daphnia magna (Water flea): 1,700 mg/l

Components

Toxicity to algae : Sodium Xylenesulfonate
96 h EC50: 230 mg/l

2-butoxyethanol
72 h EC50: 911 mg/l

Sodium silicate
72 h EC50 Desmodesmus subspicatus (green algae): 207 mg/l

Persistence and degradability

Poorly biodegradable

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

Product AS SOLD

- Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (CN_DG)

- UN number : 3266
- Description of the goods : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S.
(silicic acid, sodium salt, Potassium hydroxide)
- Class : 8
- Packing group : II
- Environmentally hazardous : No

Sea transport (IMDG/IMO)

- UN number : 3266
- Proper shipping name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S.
(silicic acid, sodium salt, Potassium hydroxide)
- Class : 8
- Packing group : II
- Marine pollutant : No

Section: 15. REGULATORY INFORMATION

National Regulations

Authored with regard to the Factories & Industrial Undertakings (Dangerous Substances) Regulations.
The components of this product are reported in the following inventories:

United States TSCA Inventory :
not determined

Canadian Domestic Substances List (DSL) :
This product contains one or several components that are not on the Canadian DSL nor NDSL.

Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :

MATERIAL SAFETY DATA SHEET

REDI-KLEEN II

not determined

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :
not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :
On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :
not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
not determined

China Inventory of Existing Chemical Substances :
On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :
not determined

Section: 16. OTHER INFORMATION

Issuing date	: 2021.09.24
Date of first issue	: 2019.08.12
version	: 1.1
Prepared by	: Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.